

2.4. DC Circuits & Kirchhoff's Laws

Imagine you're designing a camera flash. What components would you need to quickly store and release a large amount of electrical energy, and how would you control the timing?

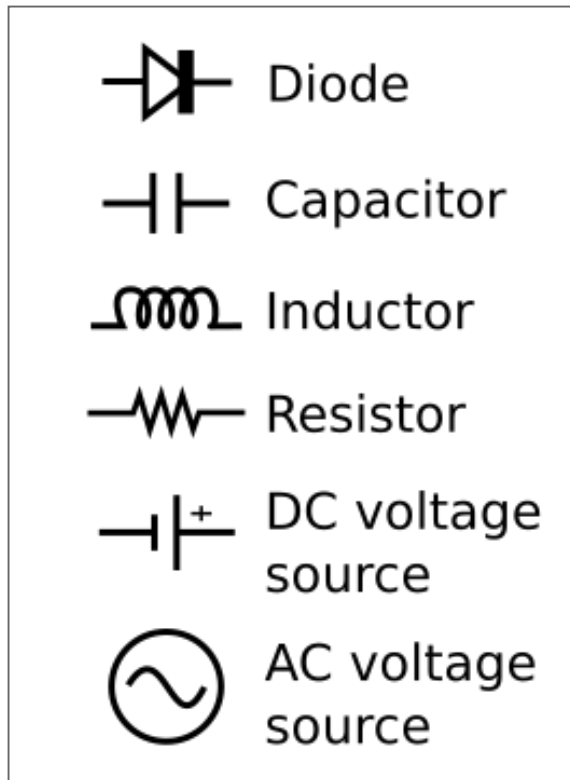
- need to design a circuit with at least
 - voltage source
 - capacitor
 - resistor
 - switch



Circuits 101

- **Electric circuit:** Elements (batteries, resistors, capacitors) connected by wires
- **Circuit states:**
 - Open: No current flow
 - Closed: Current flow
 - Short: Excessive current (hazard)
- **Voltage source (V):** Provides potential difference, drives current (I)
- **Ground:** Reference potential (e.g., Earth)
- **Voltage drop:** Potential difference across an element
- **Element connections:** Series or parallel
- **Equivalent circuits:** Simplification by combining elements into e.g. an effective resistance

Circuits 101 (cont'd)



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Open-circuit voltage of a DC source

- **Source:** Energy conversion to electrical potential
- **Open-circuit voltage** V_0 : Voltage without current flow
- **Internal resistance** R_{int} : Real sources have internal resistance
- **Voltage with current:**

$$V = V_0 - IR_{int}$$

- **Non-constant** R_{int} : Leads to non-linear current-voltage characteristics

Note: You may find electromotive force (EMF) instead of V_0 , but there is no force. Thus, this is misleading and we stick with open-circuit voltage V_0 which is equivalent to EMF if there are no leaking currents within the voltage source

Measuring Voltages and Currents

e136 & e128

- **Ammeter:** Measures current (series connection)
- **Voltmeter:** Measures voltage (parallel connection)
- **Ohmmeter:** Measures resistance
- **Multimeter:** Combines ammeter, voltmeter, ohmmeter
- **Internal resistance:**
 - Voltmeter: High (limits current)
 - Ammeter: Low (limits voltage drop)
- **Simultaneous measurement:** Configuration choice depends on resistor and device internal resistances

Resistors in series

e141 & e110

$$V = \sum V_i = \sum IR_i = I \sum R_i = IR_{eq}$$

- equivalent resistance: $R_{eq} = \sum R_i$
- voltage drops add up
- **interpretation for equivalent resistor:** increase effective length
- special case / circuit element: **potentiometer**

Resistors in parallel

e141

$$I = \sum I_i \quad \text{with} \quad \frac{V}{R_{eq}} = \sum \frac{V}{R_i} = V \frac{1}{R_i}$$

- equivalent resistance: $\frac{1}{R_{eq}} = \sum \frac{1}{R_i}$
- current splits
- net resistance is smaller than the smallest individual resistance
- **interpretation for equivalent resistor:** increase effective cross-sectional area

Capacitors in parallel

es23

$$Q_i = VC_i \quad \Rightarrow \quad Q = V \sum C_i = VC_{eq}$$

- equivalent capacitance: $C_{eq} = \sum C_i$
- voltage is the same
- total charge adds up
- **interpretation for equivalent capacitor:** increase effective plate area

Capacitors in series

es23

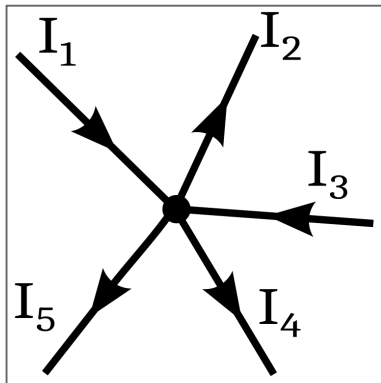
$$V_i = \frac{Q}{C_i} \quad \Rightarrow \quad V = \sum V_i = Q \sum \frac{1}{C_i} = \frac{Q}{C_{eq}}$$

- equivalent capacitance: $\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$
- same charge Q flows onto each capacitor
- voltage drops add up
- net capacitance is smaller than the smallest individual capacitance
- **interpretation for equivalent capacitor:** increase effective plate distance

Kirchhoff's junction rule

$$\sum_i I_i = 0 \quad \Leftrightarrow \quad \sum_j I_{in,j} = \sum_k I_{out,k}$$

→ **current conservation**

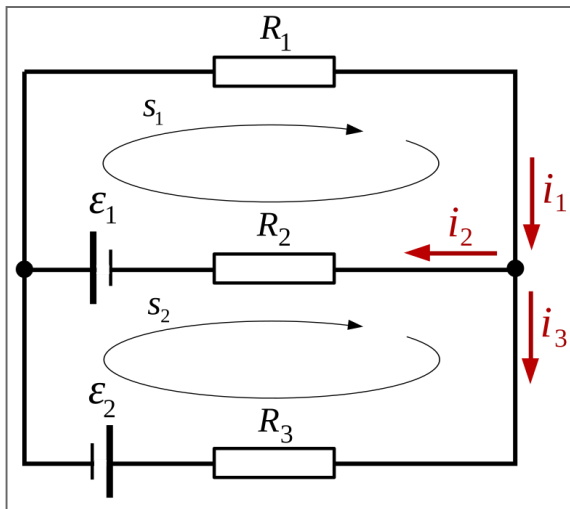


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Kirchhoff's loop rule

$$\sum_i V_i = 0$$

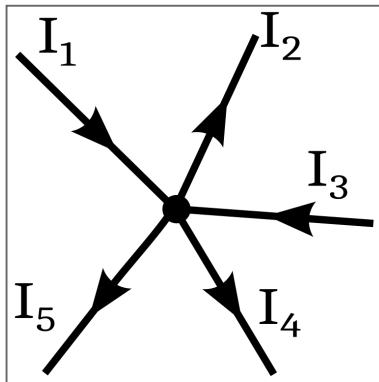
→ **energy conservation**

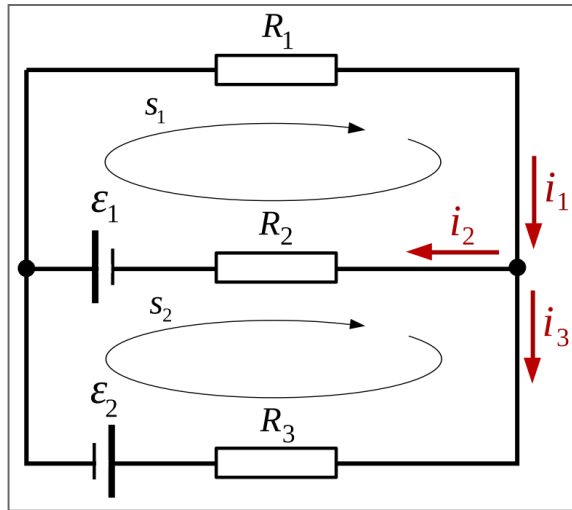


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Circuit analysis steps

1. Label currents
2. Identify unknowns
3. Apply junction rule
4. Apply loop rule
5. Solve equations
6. Vector/matrix form of equation

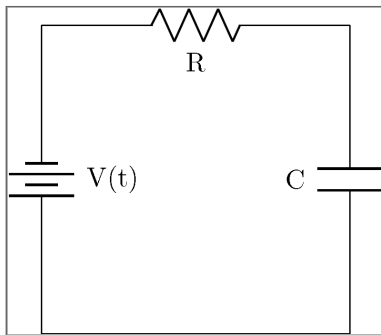




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RC-circuit

- fundamental circuit for:
 - creating precise time delays → camera flash (charging of a capacitor determines the flash duration)
 - audio filtering → smooths out unwanted fluctuations
 - simple oscillators → periodic waveforms



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RC-circuit: Charging

- **Circuit:** Resistor, capacitor, voltage source in series
- **Kirchhoff's loop rule:**

$$V_0 = V_R + V_C = IR + \frac{Q}{C}$$

- **Differential equation:** ($I = \frac{dQ}{dt}$)

$$V_0 = R \frac{dQ}{dt} + \frac{Q}{C}$$

$$\frac{dt}{RC} = \frac{dQ}{V_0 C - Q}$$

$$\frac{1}{RC} \int_0^t dt = \int_0^Q \frac{dQ}{V_0 C - Q}$$

RC-circuit: Charging (cont'd)

- **Differential equation:** ($I = \frac{dQ}{dt}$)

$$\frac{1}{RC} \int_0^t dt = \int_0^Q \frac{dQ}{V_0 C - Q}$$

- **Solution:**

$$Q(t) = Q_0(1 - e^{-t/RC})$$

$$V_C(t) = V_0(1 - e^{-t/RC})$$

$$I(t) = \frac{V_0}{R} e^{-t/RC}$$

- → At capacitor, **increasing charge & voltage** over time
- → In circuit, **decaying current** and, eventually, **no voltage drop over resistor**

- **Time constant ($\tau = RC$):**
 - Time for V_C to reach 63% of V_0
 - Time for I to decay to 37% of initial value
 - Charging is considered complete after 5 τ

RC-circuit: Discharging

- **Circuit:** Capacitor and resistor in series; **no voltage source**
- **Kirchhoff's Loop Rule:**

$$0 = V_R + V_C \quad \Rightarrow \quad IR = -\frac{Q}{C}$$

- **Differential Equation with $I = \frac{dQ}{dt}$:**

$$\frac{dQ}{dt}R = -\frac{Q}{C}$$

$$\frac{dQ}{Q} = -\frac{dt}{RC}$$

$$\int_{Q_0}^Q \frac{1}{Q} dQ = -\frac{1}{RC} \int_0^t dt$$

RC-circuit: Discharging (cont'd)

- **Differential Equation:**

$$\int_{Q_0}^Q \frac{1}{Q} dQ = -\frac{1}{RC} \int_0^t dt$$

- **Solution:**

$$Q(t) = Q_0 e^{-t/RC}$$

$$V_C(t) = V_0 e^{-t/RC}$$

$$I(t) = -I_0 e^{-t/RC}$$

- → Capacitor voltage decays exponentially
- → Current has same decay but opposite sign

RC-circuit: Periodically switching

sim RC-circuit & e111

- **Sawtooth Voltage:** Created by periodic charging and discharging
- **Practical Implementation:** Voltage source, capacitor, two resistors, and a tube
- **Charging Loop:** Capacitor charges through one resistor
- **Discharging Loop:** Tube conducts at breakdown voltage, discharging capacitor through the other resistor
- **Repeat Cycle:** Sawtooth waveform generated
- **Application:** oscillators, (heart) pacemakers, wipers, synthesizers